Integration & Applications

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Area under a curve**

Find the area under $y = x^2$ from $x = 0$ to $x = 3$, using a definite integral.

Area $= \int_0^3 x^2\,dx = \left[\frac{x^3}{3}\right]_0^3 = \frac{27}{3} = 9.$

2 **[Foundational]** • **Target: Basic antiderivatives (the power rule)**

Find $\int x^4\,dx$.

$\frac{x^5}{5} + C.$

3 **[Foundational]** • **Target: Basic antiderivatives (the power rule)**

Find $\int 3x^2\,dx$.

$x^3 + C.$

4 **[Foundational]** • **Target: Basic antiderivatives (the power rule)**

Find $\int \sqrt{x}\,dx$.

$\int x^{1/2}\,dx = \frac{2}{3}x^{3/2} + C.$

5 **[Foundational]** • **Target: Basic antiderivatives (the power rule)**

Find $\int \frac{1}{x^3}\,dx$.

$\int x^{-3}\,dx = -\frac{1}{2}x^{-2} + C = -\frac{1}{2x^2} + C.$

6 **[Foundational]** • **Target: Antiderivatives of polynomials**

Find $\int (4x^3 - 6x + 5)\,dx$.

$x^4 - 3x^2 + 5x + C.$

7 **[Foundational]** • **Target: Antiderivatives of polynomials**

Find $\int (2x + 7)\,dx$.

$x^2 + 7x + C.$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Antiderivatives of polynomials

Find $\int (x^2 - 4x + 3)\,dx$.

$\frac{x^3}{3} - 2x^2 + 3x + C$.

9 [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int \cos x\,dx$.

$\sin x + C$.

10 [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int \sin x\,dx$.

$-\cos x + C$.

11 [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int e^x\,dx$.

$e^x + C$.

12 [Intermediate] • Target: Evaluating definite integrals

Evaluate $\int_0^2 3x^2\,dx$.

$[x^3]_0^2 = 8$.

13 [Intermediate] • Target: Evaluating definite integrals

Evaluate $\int_1^3 2x\,dx$.

$[x^2]_1^3 = 9 - 1 = 8$.

14 [Intermediate] • Target: Evaluating definite integrals

Evaluate $\int_0^4 x\,dx$.

$\left[\frac{x^2}{2}\right]_0^4 = 8$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Evaluating definite integrals • Cross-Lesson: 4.01 Trigonometry

Evaluate $\int_{-1}^2 3x^2\,dx$.

$[x^3]_{-1}^2 = 8 - (-1) = 9$.

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16 [Challenging] • Target: Definite integrals with polynomial integrands • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Evaluate $\int_0^2 (x^2 + 1)\,dx$.

$\left[\frac{x^3}{3} + x\right]_0^2 = \frac{8}{3} + 2 = \frac{14}{3}$.

17 [Challenging] • Target: Definite integrals with polynomial integrands • Cross-Lesson: 6.01 Limits & Derivatives

Evaluate $\int_1^3 (2x^2 - 3x + 1)\,dx$.

$F(x) = \frac{2x^3}{3} - \frac{3x^2}{2} + x$. $F(3) = 18 - 13.5 + 3 = 7.5$. $F(1) = 0.667 - 1.5 + 1 = 0.167$. Integral $= 7.5 - 0.167 = \frac{22}{3}$.

18 [Challenging] • Target: Definite integrals with polynomial integrands • Cross-Lesson: 8.01 Exponentials & Logarithms

Evaluate $\int_0^1 (6x^2 - 4x + 2)\,dx$.

$[2x^3 - 2x^2 + 2x]_0^1 = (2 - 2 + 2) - 0 = 2$.

19 [Challenging] • Target: Definite integrals with polynomial integrands • Cross-Lesson: 9.01 Statistics & Probability

Evaluate $\int_{-2}^1 (x^2 - x)\,dx$.

$F(x) = \frac{x^3}{3} - \frac{x^2}{2}$. $F(1) = -\frac{1}{6}$. $F(-2) = -\frac{8}{3} - 2 = -\frac{14}{3}$. Integral $= -\frac{1}{6} + \frac{14}{3} = \frac{9}{2}$.

20 [Challenging] • Target: Definite integrals of trigonometric and exponential functions • Cross-Lesson: 1.01 Algebra & Equations

Evaluate $\int_0^{\pi/2} \cos x\,dx$.

$[\sin x]_0^{\pi/2} = 1 - 0 = 1$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Area under a curve

The graph shows $y = x^2$. Find the area under the curve from $x = 0$ to $x = 3$, using a definite integral.

a. 9 ✓

b. 18

c. 3

d. 27

$\left[\frac{x^3}{3}\right]_0^3 = 9$.

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22 [Foundational] • Target: MCQ — Area under a curve

The graph shows $y = 4 - x^2$. Find the area under the curve from $x = -2$ to $x = 2$.

a. $\frac{16}{3}$

b. 32

c. 16

d. $\frac{32}{3}$ ✓

$\left[4x - \frac{x^3}{3}\right]_{-2}^{2} = \frac{16}{3} - \left(-\frac{16}{3}\right) = \frac{32}{3}$.

23 [Foundational] • Target: MCQ — Word problems: integration in context

The graph shows a particle's velocity $v(t) = 8t - 2t^2$ for $0 \le t \le 4$. Find the particle's displacement (the area under the curve).

a. $\frac{64}{3}$ ✓

b. 16

c. 32

d. 64

$\int_0^4 (8t - 2t^2)\,dt = \frac{64}{3}$ m.

24 [Foundational] • Target: MCQ — Basic antiderivatives (the power rule)

Find $\int x^6\,dx$.

a. $\frac{x^6}{7} + C$

b. $7x^5 + C$

c. $\frac{x^7}{6} + C$

d. $\frac{x^7}{7} + C$ ✓

Power rule: $\frac{x^{6+1}}{6+1} + C$.

25 [Foundational] • Target: MCQ — Basic antiderivatives (the power rule)

Find $\int 7\,dx$.

a. C

b. $7x + C$ ✓

c. $7 + C$

d. $\frac{7x^2}{2} + C$

The antiderivative of a constant k is $kx + C$.

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26 [Foundational] • Target: MCQ — Antiderivatives of polynomials

Find $\int (x^2 - 6x + 5)\,dx$.

a. $\frac{x^3}{3} - 3x^2 + 5x + C$ ✓

b. $\frac{x^3}{3} - 6x^2 + 5x + C$

c. $\frac{x^3}{3} - 3x^2 + C$

d. $x^3 - 3x^2 + 5x + C$

Integrate each term.

27 [Foundational] • Target: MCQ — Antiderivatives of trigonometric and exponential functions

Find $\int (e^x + 2)\,dx$.

a. $e^x + 2 + C$

b. $e^x + 2x + C$ ✓

c. $xe^x + 2x + C$

d. $e^x + C$

$\int e^x\,dx = e^x$; $\int 2\,dx = 2x$.

28 [Foundational] • Target: MCQ — Evaluating definite integrals

Evaluate $\int_0^5 x\,dx$.

a. 5

b. 12.5 ✓

c. 25

d. 10

$\left[\frac{x^2}{2}\right]_0^5 = 12.5$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Definite integrals with polynomial integrands

Evaluate $\int_1^4 (3x^2 - 2x + 1)\,dx$.

a. 50

b. 52

c. 63

d. 51 ✓

$[x^3 - x^2 + x]_1^4 = 52 - 1 = 51$.

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30 [Intermediate] • Target: MCQ — Definite integrals of trigonometric and exponential functions

Evaluate $\int_0^{\pi} \cos x \, dx$.

a. π

b. -2

c. 0 ✓

d. 2

$[\sin x]_0^{\pi} = 0 - 0 = 0$.

31 [Intermediate] • Target: MCQ — Signed area and regions below the axis

Evaluate $\int_0^3 (x - 3) \, dx$ and explain what the sign of the result means.

a. 4.5; the region lies above the x-axis

b. -4.5; the region lies above the x-axis

c. 4.5; the region lies below the x-axis

d. -4.5**; the region lies below the** x**-axis** ✓

$\left[\frac{x^2}{2} - 3x\right]_0^3 = 4.5 - 9 = -4.5$; negative means the region lies below the x-axis.

32 [Intermediate] • Target: MCQ — Properties of definite integrals

Given $\int_0^3 f(x) \, dx = 6$ and $\int_3^7 f(x) \, dx = 9$, find $\int_0^7 f(x) \, dx$.

a. 13

b. 3

c. 15 ✓

d. 54

$6 + 9 = 15$.

33 [Intermediate] • Target: MCQ — Finding a function from its derivative

A function satisfies $f'(x) = 4x^2 + 3x$ and $f(0) = 2$. Find $f(x)$.

a. $\frac{4x^3}{3} + 3x^2 + 2$

b. $\frac{4x^3}{3} + 1.5x^2$

c. $\frac{4x^3}{3} + 1.5x^2 + 2$ ✓

d. $4x^3 + 1.5x^2 + 2$

$f(x) = \frac{4x^3}{3} + 1.5x^2 + C$; $f(0) = C = 2$.

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34 [Intermediate] • Target: MCQ — Average value of a function

Find the average value of $f(x) = 3x + 2$ on $[0, 4]$.

a. 32

b. 4

c. 14

d. 8 ✓

$\frac{1}{4}[1.5x^2 + 2x]_0^4 = \frac{1}{4}(32) = 8$.

35 [Intermediate] • Target: MCQ — Word problems: integration in context

A factory's production rate is $p(t) = 50 + 4t$ items per hour, for $0 \le t \le 8$. Find the total number of items produced during this 8-hour period.

a. 528 ✓

b. 656

c. 450

d. 400

$\int_0^8 (50 + 4t)\,dt = [50t + 2t^2]_0^8 = 528$ items.

36 [Intermediate] • Target: MCQ — Evaluating definite integrals

Find $\int -x^2\,dx$.

a. $-\frac{1}{3}x^3 + C$ ✓

b. $-\frac{1}{4}x^3 + C$

c. $-\frac{1}{2}x^2 + C$

d. $-x^3 + C$

$-\frac{1}{3}x^3 + C$.

37 [Intermediate] • Target: MCQ — Definite integrals with polynomial integrands

Find $\int (4x + 9)\,dx$.

a. $2x^2 + 9x + C$ ✓

b. $2x^2 - 9x + C$

c. $4x + 9 + C$

d. $2x^2 + 11x + C$

$2x^2 + 9x + C$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: MCQ — Definite integrals of trigonometric and exponential functions • Cross-Lesson: 8.01 Exponentials & Logarithms

Find $\int 3\cos x\,dx$.

a. $3-\sin x+C$

b. $4\sin x+C$

c. $3\sin x+C$ ✓

d. $3\cos x+C$

$3\sin x+C$.

39 [Challenging] • Target: MCQ — Area under a curve • Cross-Lesson: 9.01 Statistics & Probability

Evaluate $\int_1^3 6x\,dx$.

a. 22

b. 26

c. 24 ✓

d. -24

$[3x^2]_1^3=27-3=24$.

40 [Challenging] • Target: MCQ — Signed area and regions below the axis • Cross-Lesson: 1.01 Algebra & Equations

Evaluate $\int_0^3 (9x^2-1)\,dx$.

a. 76

b. 78 ✓

c. 80

d. -78

$[3x^3-x]_0^3=78$.

41 [Challenging] • Target: MCQ — Properties of definite integrals • Cross-Lesson: 2.01 Functions & Transformations

A function satisfies $f'(x)=-2x+5$ and $f(1)=10$. Find $f(x)$.

a. $f(x)=x^2+5x+6$

b. $f(x)=-x^2+5x+9$

c. $f(x)=-x^2+5x-6$

d. $f(x)=-x^2+5x+6$ ✓

$f(x)=-x^2+5x+C$. $f(1)=4+C=10\Rightarrow C=6$. $f(x)=-x^2+5x+6$.

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42 [Challenging] • Target: MCQ — Finding a function from its derivative • Cross-Lesson: 3.01 Sequences & Series

Find the area under $y = 3x^2$ from $x = 0$ to $x = 4$, using a definite integral.

a. 192

b. 66

c. 62

d. 64 ✓

Area $= \int_0^4 3x^2\,dx = \left[\frac{3x^3}{3}\right]_0^4 = \frac{192}{3} = 64$.

43 [Challenging] • Target: MCQ — Average value of a function • Cross-Lesson: 4.01 Trigonometry

Given $\int_0^2 f(x)\,dx = 5$ and $\int_2^6 f(x)\,dx = 2$, find $\int_0^6 f(x)\,dx$.

a. 3

b. 9

c. 10

d. 7 ✓

By additivity over intervals: $5 + 2 = 7$.

44 [Challenging] • Target: MCQ — Word problems: integration in context • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Given $\int_0^5 f(x)\,dx = 3$, find $\int_5^0 f(x)\,dx$.

a. 3

b. -3 ✓

c. -5

d. 5

Reversing the limits negates the integral: $\int_5^0 f(x)\,dx = -3$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Limits & Derivatives

Find the average value of $f(x) = -2x + 4$ on $[1, 4]$.

a. -1 ✓

b. -2

c. 0

d. 1

$\int_1^4 (-2x + 4)\,dx = \left[\frac{-2x^2}{2} + 4x\right]_1^4 = -3$. avg $= \frac{-3}{4-1} = -1$.

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